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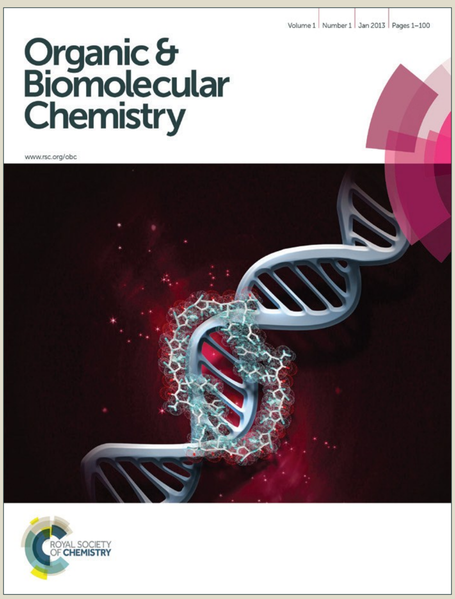

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ARTICLE

Stereoselective Synthesis and Antitumoral Activity of *Z*-Enyne Pseudoglycosides

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An efficient approach for the synthesis of *Z*-1,3-enynes based on the coupling reaction of *Z*-vinyl tellurides and alkynes containing a pseudoglycoside moiety is described. The products were obtained in good yields via a stereoselective way. Preliminary screening against three tumor cell lineages indicated that the synthesized compounds are promissing intermediates for the synthesis of an array of more potent target structures.

Introduction

The enediyne and enyne motifs are structural units present in several natural products which exhibits a variety of biological activities. Examples are (+)-Phorbaside[1] and neocarzinostatin chromophore[2] (Figure 1).

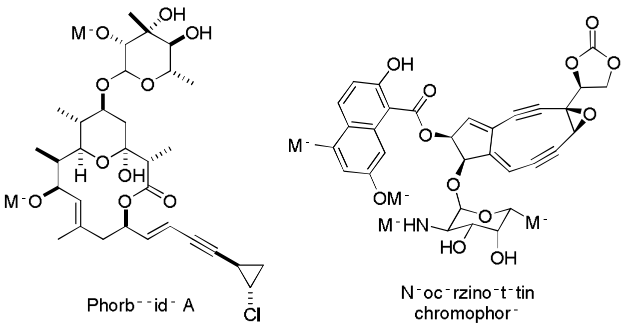

Figure 1. Examples of highly unsaturated natural products

The practical use of these compounds is sometimes limited to their toxicity and scarcity from natural sources. However, in a simpler view, they consist structurally in an aglycone with a stereodefined enyne and a sugar residue connected by an *O*-glycosidic bond. Thus, the synthesis of analogues with lower structural complexity that could mimic the mode of action of these compounds would be of the great interest. This new approach would combine the structural features of different classes of compounds to provide new compounds with improved or unprecedent biological activities in a short span.[3]

Several methods can be employed for the synthesis of stereodefined enynes, being the Pd-catalysed Sonogashira coupling the most common.[4] Other approaches based on variations of the Sonogashira reaction using less expensive metals such as Fe,[5] Cu[6] and Ni[7] were also described.

Despite the advances obtained for this reaction, the Sonogashira coupling requires the stereoselective preparation of the appropriate alkenyl halide, which sometimes demands additional steps or expensive chemicals.[8] In addition, some developments in the Sonogashira coupling require the use of pyrophoric phophines and the competing Glaser type oxidative dimerization of the alkyne is still a problem.[9]

An alternative and efficient method for the stereoselective synthesis of alkenes is the hydrotelluration reaction.[10] The method differs from the other hydrometallation reactions, such as hydroalumination,[11] hydroboration[12] and hydrozirconation[13] because it proceeds through the *anti* addition of an organotellurolate anion to the appropriate alkyne to yield a *Z*-alkene as a major product.

Results and Discussion

In an initial approach, alkynes **1a**-**d** were submitted to the hydrotelluration conditions to yield the corresponding *Z*-vinyl tellurides **2a**-**d** in good yield in a very stereo- and regioselective way (Table 1). The only exception occurred when TIPS protected propargyl alcohol was used, where a 90:10 mixture of two regioisomers **2d** and **3d** was observed (Table 1, entry 4). The vinyl tellurides **2d** and **3d**, however, could be easily separated by flash column chromatography and the regioisomeric purity was determined by ^{1}H NMR and confirmed by ^{125}Te NMR and gas chromatography.[14]

Next, 3,4,6-tri-*O*-acetyl-D-glucal, **4**[15] was used as a glycosyl donor in a reaction using different alcohols catalyzed by tellurium tetrachloride (IV). The corresponding pseudoglycosides **5** were obtained in good yields and anomeric selectivities.[16] The results are depicted on Table 2. Compounds **5a**-**d** were then purified by chromatographic column to yield pure α-anomer and the stereochemistry was supported by a NOESY experiment.[17]

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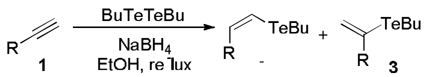
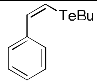
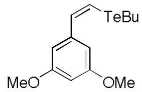
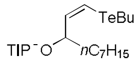
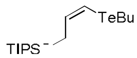
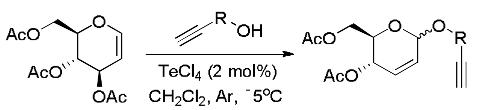
Table 1. Stereoselective synthesis of vinyl tellurides[a]

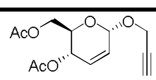
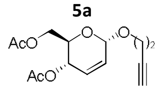
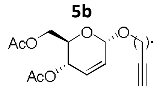
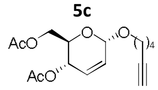
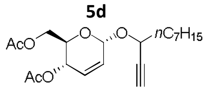


entry	R	Time (h)	Vinyl Telluride	Yield(%)[b] (**2**:**3**)[c]
1	C_6H_5 **1a**	5.0	 **2a**	91 (100:0)
2	3,5-OMe-C_6H_3 **1b**	5.0	 **2b**	85 (100:0)
3	TIPSO(CH)*n*-C_7H_{15} **1c**	3.0	 **2c**	90 (100:0)
4	TIPSOCH$_2$ **1d**	5.0	 **2d**	89 (90:10)

[a]*Reaction conditions*: Reactions were performed with the appropriate alkyne **1a**-**d** (5 mmol), dibutylditelluride (2.5 mmol) and sodium borohydride (7 mmol) in ethanol (40 mL) under reflux for the time indicated. [b]Isolated yield; [c]The regioisomeric ratios were obtained by ^{1}H NMR and confirmed by ^{125}Te NMR and gas chromatography.

Table 2. Synthesis of pseudoglycosides catalyzed by $TeCl_4$[a]



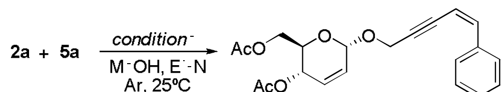
entry	R	Time (min)	5	Yield(%)[b] (α:β)[c]
1	CH$_2$	3.0	 **5a**	92 (89:11)
2	(CH$_2$)$_2$	3.0	 **5b**	91 (90:10)
3	(CH$_2$)$_3$	3.0	 **5c**	86 (88:12)
4	(CH$_2$)$_4$	3.0	 **5d**	85 (86:14)
5	CH(C_7H_{17})	5.0	 **5e**	89 (90:10)

[a]*Reaction conditions*: Reactions were performed with **4** (2 mmol), the appropriate alkyne (2.4 mmol) and $TeCl_4$ (11 mg, 2 mol%) in CH_2Cl_2 (15 mL) at 25°C for the time indicated. [b]Isolated yield; [c]The anomeric ratios were obtained by ^{1}H NMR and confirmed by gas chromatography.

The vinyl tellurides **2** and α-pseudoglycosides **5** were then submitted to a palladium-catalyzed cross-coupling reaction.[18] For preliminary optimization of the reaction conditions, **2a** and α-**5a** were used as model compounds and treated with a variety of catalysts using Et_3N as a base and methanol as solvent at room temperature. The results are depicted on Table 3.

From Table 3 it can be seen that when $PdCl_2$ and CuI were used as the catalytic system in an amount of 5 mol% **6a** was obtained in low yield (Table 3, entry 1). When the amount of catalyst was increased to 20 and 40 mol%, respectively, higher yields were observed together with a decrease in the reaction time (Table 3, entries 2 and 3). The replacement of $PdCl_2$ for the less expensive $Pd(OAc)_2$ in the presence or absence of CuI did not gave the desired product (Table 3, entries 4 and 5). When the reaction was performed in the absence of CuI using only $PdCl_2$ as catalyst, **6a** was obtained in 48% yield (Table 3, entry 6). Likewise, in the absence of $PdCl_2$ and using only CuI as catalyst, the formation of **6a** was not observed after 6h (Table 3, entry 7).

Table 3. Effect of reaction conditions for the coupling reaction[a]



entry	conditions	Time (h)	**6a** (%)[b]
1	$PdCl_2$ (5 mol%), CuI (5 mol%)	6.0	22
2	$PdCl_2$ (20 mol%), CuI (20 mol%)	0.75	89
3	$PdCl_2$ (40 mol%), CuI (40 mol%)	0.5	87
4	$Pd(OAc)_2$ (20 mol%),	6.0	0
5	$Pd(OAc)_2$ (20 mol%), CuI (20 mol%)	6.0	0
6	$PdCl_2$ (20 mol%)	6.0	48
7	CuI (40 mol%)	6.0	0

[a]*Reaction conditions*: Reactions were performed with **2a** (0.5 mmol), **5a** (0.55 mmol) in MeOH (5 mL) and Et_3N (0.3 mL) at 25°C using the conditions and time indicated. [b]Isolated yield; [c]The anomeric purity of the obtained compounds was confirmed by ^{1}H NMR and gas chromatography.

With the optimized reaction conditions the scope of the cross-coupling reaction was extended to different substrates and the results are described on Table 4.

The method is robust and the reaction seems not to be sensitive to the type of functional group present in the starting telluride since in all cases good yields were observed without isomerization of the *Z*-double bond.

The increment in the distance of the glycosyl moiety from the triple bond did not affect the yield considerably (Table 4, entries 1-4). The reaction also does not seem to be affected by the type of substituent present on the aromatic ring once the coupling reaction of the vinyl telluride **2b** and α-pseudoglycoside **5a** led to the desired product in good yield (Table 4, entry 5).

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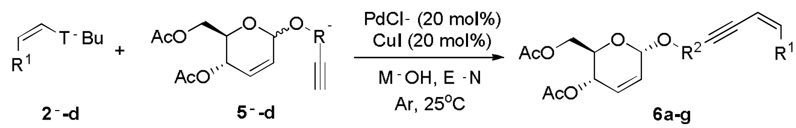
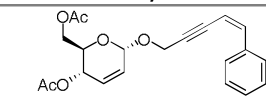
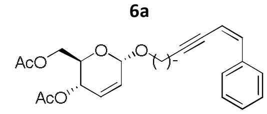
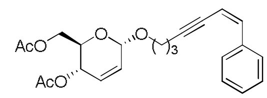
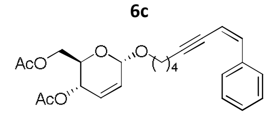
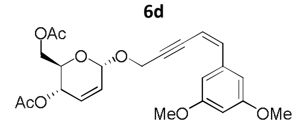
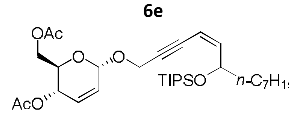
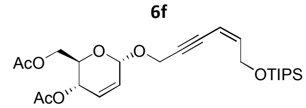
The coupling reaction of *Z*-vinyl telluride **2c** and α-pseudoglycoside **5a** also gave the corresponding enyne **6f** in good yield (Table 4, entry 6). It is interesting to note that enyne **6f** is the core of Seselidiol, a natural product isolated from the roots of *Seseli mairei Wolff* (Umbelliferae) which has showed significant cytotoxicity against some tumor cells.[19]

Table 4. Scope and generality for the coupling reaction of *Z*-vinyl tellurides and α-pseudoglycosides[a]

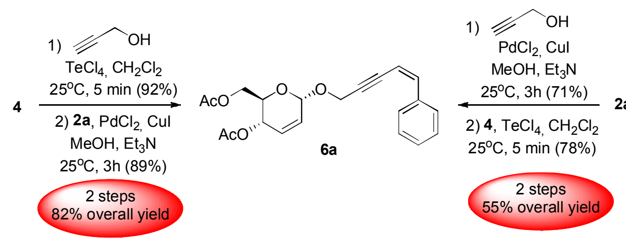


entry	Telluride	Pseudoglycoside	Time(h)	enyne	Yield (%)[b]
1	**2a**	**5a**	0.75	 **6a**	89
2	**2a**	**5b**	1.0	 **6b**	86
3	**2a**	**5c**	1.0	 **6c**	87
4	**2a**	**5d**	1.0	 **6d**	85
5	**2b**	**5a**	0.5	 **6e**	85
6	**2c**	**5a**	1.0	 **6f**	85
7	**2d**	**5a**	1.0	 **6g**	84

[a]*Reaction conditions*: Reactions were performed with **2a-d** (0.5 mmol), **5a-d** (0.55 mmol) in MeOH (5 mL) and Et_3N (0.3 mL) at 25°C using the conditions and time indicated. [b]Isolated yield.

Evidently one of the main objectives in the development of new synthetic methodologies for biologic active compounds is to obtain the products in high yields and purity. Thus, two routes were compared in order to stablish which one would give better overall yield (Scheme 1).

In the first approach, glycosidation of propargyl alcohol using 3,4,6-tri-O-acetyl-*D*-glucal, **4** catalyzed by tellurium tetrachloride (IV) gave the corresponding pseudoglycoside **5a** in 92% yield. Subsequent coupling reaction with vinyl telluride **2a** gave the corresponding *Z*-enyne in 89% yield. Using this sequence the overall yield for **6a** was 82% after two steps (Scheme 1).



Scheme 1. Overall yield for the synthesis of Z-enyne compounds using two different routes

Conversely, vinyl telluride **2a** (Table 1, entry 1) was submitted to the coupling reaction with propargyl alcohol to yield the corresponding *Z*-enyne in 71% yield. Further glycosidation using 3,4,6-tri-O-acetyl-*D*-glucal, **4** as a glycosyl

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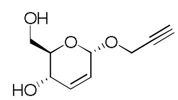
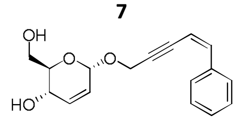
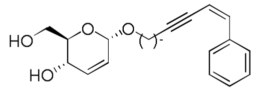
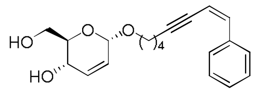
donor catalyzed by tellurium tetrachloride (IV) gave the corresponding pseudoglycoside **6a** in 78% yield. Using this sequence, the overall yield for **6a** was 55% after two steps (Scheme 1)

The antitumoral activity of some enyne and enediyne natural products has been previously described. These facts prompt us to submit the synthesized compounds to the evaluation of their cytotoxic effects on some tumor cells. The synthesized compounds were then submitted to a preliminary screening at 25 μg.mL^{-1} against HL-60 (human acute promyelocytic leukemia), NCI-H292 (human lung carcinoma) and MCF-7 (human breast adenocarcinoma) tumor cells lineages using the MTT assay.[20] Generally, the samples with growth inhibition over 90% are used to determine the IC_{50} values (concentration that causes 50% growth inhibition).

Table 5. Preliminary screening at 25 μg.mL^{-1} for the synthesized compounds against three tumor cells lineages.

entry	compound	HL-60	NCI-H292	MCF-7
1	**6a**	88.3±0.5	44.8±5.8	28.4±3.0
2	**6b**	86,3±0.6	69.2±6.4	48.7±5.6
3	**6c**	99.8±0.1	62.76±5.0	72.8±0.3
4	**6d**	98.7±0.1	57.5±4.2	63.6±0.6
5	**6e**	75.0±2.6	28.0±2.5	34.3±0.2
6	**6f**	61.1±14,3	42.1±0.25	-
7	**5a**	5.5±5.6	41.0±7.0	10.7±0.2
8	 **7**	28.4±3.3	42.8±8.8	17.5±5.0
9	 **8**	54.2±2.5	35.4±2.0	15.7±6.35
10	 **9**	90.8±0.7	90.1±0.1	89.0±1.3
11	 **10**	95.7±0.8	90.0±0.1	90.8±2.0
DOX[a]	-	98.1±0.8	94.9±0.2	80.3±0,7

[a] Doxorubicin 5 μg.mL^{-1} was used as positive control.

From Table 5 it can be observed that when the distance of the glycoside portion from the triple bond was increased, higher values of antiproliferative activity were observed (Table 5, entries 1-4) being the best results observed for compounds **6c** (Table 5, entry 3) and **6d** (Table 5, entry 4) against human promyelocytic leukemia cells, demonstrating the potential of the newly synthesized compounds as antitumoral agents. Compounds **6e** and **6f** exhibited only moderate antiproliferative activity against all tested cancer cell lines (Table 5, entries 5 and 6).

In order to explore which part of the tested compounds, the *Z*-enyne or the α-pseudoglycoside moiety, would be responsible for the antiproliferative activity, α-pseudoglycoside **5a** (Table 5, entry 8) was evaluated in order to be compared to *Z*-enyne **6a** (Table 5, entry 1). Lower antiproliferative activities for all tested cancer cell lines were observed.

A major problem on the development and screening of new chemotherapeutic drugs is the lower solubility of them in aqueous medium. To circumvent this problem, high concentrations of surfactants and (or) co-solvents are generally required.[21] Thus, in order to increase the solubility in aqueous medium, the acetyl groups present in some of the screened compounds were hydrolysed using potassium carbonate in an aqueous solution of methanol to give the corresponding products **7**-**10** in good yields.[22]

Low antiproliferative activities were observed for compounds **7** and **8** (Table 5, entries 8 and 9). However,

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compounds **9** and **10** displayed good cytotoxic activities against all tested cell lines (Table 5, entries 10 and 11).

The compounds that shown growth inhibition over 90% were then used to determine the IC_{50} values. The results are described on Table 6.

Table 6. Cytotoxicity of compounds against tumor cell lines

entry		IC_{50} (μg.mL^{-1})		
		HL-60	**NCI-H292**	**MCF-7**
1	**6c**	6.5±2.0	-	-
2	**6d**	5.0±1.5	-	-
3	**9**	3.8±0.9	6.9±2.3	2.6±0.9
4	**10**	4.7±0.8	2.0±0.5	1.4±0.5
5	**DOX**[a]	0.34	0.02	0.03

[a] Doxorubicin was used as positive control.

Compounds **6c** and **6d** were active only against HL-60 cancer cell lines and not active against other tumor tissue cells tested, such as NCI-H292 or MCF-7 cancer cell lines (Table 6, entries 1 and 2). Noteworthy, compounds **9** and **10** showed good cytotoxicity against all tested cell lines (Table 6, entries 3 and 4) being compound **10** more active against MCF-7 cell lines. These results demonstrated that the removal of acetyl groups associated with the appropriate distance of the *Z*-enyne and α-pseudoglycoside moieties are important to improve the biological activity of these compounds once compounds **9** and **10** shown antiproliferative activities against cancer cell lines less susceptible to oxidative stress.

Conclusions

In summary, we have shown an efficient method for the synthesis of lower structural complexity analogues of enynes and enediyne natural products in a regio- and stereoselective way. The method features the use of commercially available reagents for the synthesis of precursors, and the *Z*-enyne coupled products were obtained in high yields and isomeric purity.

Some of the synthesized compounds exhibited good antitumoral activity and are promising intermediates for the synthesis of an array of more potent target structures once the *Z*-enyne and the α-pseudoglycoside moieties can be used as additional points of diversity. Additional studies will be performed to address these possibilities.

Experimental

General Information

All reactions were conducted in flame dried glassware under nitrogen. All solvents were purified before use.[23] THF was dried by distillation from sodium benzophenone ketyl. CH_2Cl_2 was dried by distillation from CaH_2. All other commercially available reagents and solvents were used as received. Reactions were monitored by thin-layer chromatography on 0.25 mm E. Merck silica gel 60 plates (F254) using UV light, vanillin and *p*-anisaldehyde as visualizing agents. Column chromatographic purification was performed using Silica Gel 60 (230-400 mesh) unless indicated otherwise. All compounds purified by chromatography were sufficiently pure for use in further experiments, unless indicated otherwise.[24]

^{1}H and ^{13}C NMR data were recorded using Varian UNITY PLUS spectrometers. ^{1}H NMR chemical shifts are reported as delta (δ) units in parts per million (ppm) relative to residual $CDCl_3$. Coupling constants (*J*) were reported in Hertz (Hz). ^{125}Te NMR data were obtained at 94.6 MHz using diphenyl ditelluride as an external reference (422.0 ppm).

Low resolution mass spectra were obtained using a Shimadzu QP-5050A Spectometer (70 eV) using helium 4.5 as a carrier gas and a DB-5 column (30m X 0,25µm). High resolution mass spectra were obtained by the São Paulo University, Chemistry Institute mass spectrometry facility, run by the electro spray ionization time-of-flight (ESI-TOF) mode on a Bruker Micro Tof Ic Bruker Daltonics mass spectrometer. The melting points (mp) were obtained using a Electrothermal 9100 melting point apparatus and are not corrected.

General Procedure for the Stereoselective Synthesis of Vinyl Tellurides (2a-d)

The appropriate alkyne **1a**-**d** (5 mmol) and dibutylditelluride (0.94 g; 2.5 mmol) were dissolved in absolute etanol (40 mL) at room temperature. Finely powdered sodium borohydride was added in small portions to the above solution until a yellowish color is observed. Additional sodium borohydride was added as necessary to maintain a yellow color (indicative of the butyltellurolate anion). The solution was heated to reflux for the time indicated on Table 1. After this period, the reaction was cooled to room temperature and quenched by the addition of a saturated solution of sodium bicarbonate (20 mL). The mixture was diluted with EtOAc (20 mL). The organic layer was isolated and washed with water (50 mL) and brine (50 mL) before drying over $MgSO_4$. The organic phase was filtered, concentrated *in vacuo* and the residue was purified by silica gel chromatography [hexanes/EtOAc (8:2)].

(*Z*)-Butyl(styryl)tellane (2a):[25] Yellow oil; 91% (1.32 g); ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.43-7.36 (m, 2H), 7.31-7.24 (m, 4H), 7.02 (d, *J* = 10.8 Hz, 1H), 2.76 (t, *J* = 7.5 Hz, 2H), 1.85 (qui, *J* = 7.5 Hz, 2H), 1.44 (sext, *J* = 7.5 Hz, 2H), 0.96 (t, *J* = 7.5 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 138.6, 136.4, 127.9, 127.2, 126.9, 104.9, 33.6, 24.6, 13.0, 8.6; ^{125}Te NMR (94.6 MHz, $CDCl_3$) δ 330.9.

(*Z*)-Butyl(3,5-dimethoxystyryl)tellane (2b): Yellow oil; 85% (1.49 g); IR (thin film cm^{-1}) 3050, 2952, 1595, 1457, 1421, 1339, 1299, 1253, 1197, 1152, 1060, 929, 838, 671; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.33 (d, *J* = 10.7 Hz, 1H); 7.01 (d, *J* = 10.7 Hz, 1H), 6.48 (d, *J* = 2.2 Hz, 2H), 6.39-6.38 (m, 1H), 3.82 (s, 6H), 2.74 (t, *J* = 7.5 Hz, 2H), 1.84 (qui, *J* = 7.2 Hz, 2H), 1.43 (sext, *J* = 7.2 Hz, 2H), 0.98 (t, *J* = 7.2 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 160.3, 140.6, 136.2, 105.8, 104.9, 99.5, 54.9, 33.5, 24.4, 12.9, 8.6; ^{125}Te NMR (94.6 MHz, $CDCl_3$) δ 336.0. HRMS (ESI, MeOH:H_2O) calcd for $C_{14}H_{20}O_2TeNa$ [M + Na]$^+$ 373.0423, found 373.0428.

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(*Z*)-(1-(Butyltellanyl)dec-1-en-3-yloxy)triisopropylsilane (2c):[26] Yellow oil; 90% (2.24 g); ^{1}H NMR (300 MHz, $CDCl_3$) δ 6.62 (d, *J* = 9.6 Hz, 1H), 6.23 (dt, *J* = 9.6 and 1.8 Hz, 1H), 4.22-4.16 (m, 1H), 2.71-2.58 (m, 2H), 1.80-1.68 (m, 2H), 1.44-1.36 (m, 2H), 1.35-1.25 (m, 12 H), 1.20-1.00 (m, 21H), 0.93 (t, *J* = 7.5 Hz, 3 H), 0.89 (t, *J* = 7.5 Hz, 3 H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 143.2, 101.1, 75.4, 37.7, 33.8, 31.5, 29.5, 28.9, 24.6, 24.5, 22.3, 17.8, 13.7, 13.0, 12.0, 6.7; ^{125}Te NMR (94.6 MHz, $CDCl_3$) δ 274.3.

(*Z*)-(3-(Butyltellanyl)allyloxy)triisopropylsilane (2d):[14] Yellow oil; 89% (1.78 g); ^{1}H NMR (300 MHz, $CDCl_3$) δ 6.69 (dt, *J* = 9.9 and 1.5 Hz, 1H), 6.38 (dt, *J* = 9.9 and 4.8 Hz, 1H), 4.21 (dd, *J* = 4.8 and 1.5 Hz, 2H), 2.60 (t, *J* = 7.5 Hz, 2H), 1.75 (qui, *J* = 7.5 Hz, 2H), 1.37 (sext, *J* = 7.5 Hz, 2H), 1.20-1.00 (m, 21H), 0.90 (t, *J* = 7.5 Hz, 3 H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 138.1, 102.1, 65.7, 34.0, 24.9, 17.9, 13.7, 11.9, 7.0; ^{125}Te NMR (94.6 MHz, $CDCl_3$) δ 298.6.

General Procedure for the synthesis of 3,4,6-tri-O-acetyl-D-glucal (4): To a 250 mL round-bottomed flask containing anhydrous *D*-glucose (3.6 g, 20 mmol) was added Ac_2O (12.2 mL, 13.1 g, 127.3 mol) and H_2SO_4 (0.05 mL, 0.09 g, 1 mmol). The mixture was placed in an ultrasound bath and sonicated for 15 min at room temperature. To the same flask was then added a 31% HBr/AcOH solution (54.5 mL) [HBr 48% (10.9 mL) in Ac_2O (43.6 mL)] and the mixture was again sonicated for 45 min. Anhydrous NaOAc (7.3 g, 88.7 mmol) was added to the flask and sonicated for 10 min. Finally, a suspension of $CuSO_4.5H_2O$ (2.2 g, 8.7 mmol) and zinc dust (36.4 g, 556 mmol) in water (36 mL) and AcOH (55 mL) containing $NaOAc.3H_2O$ (34.5 g, 254 mmol). This mixture was then sonicated at room temperature for 20 min. The solid residue was filtered off and washed with EtOAc (3 x 150 mL) and water (2 x 100 mL). The combined organic phases were washed with a saturated solution of $NaHCO_3$ (5 x 100 mL) and brine (2 x 100 mL) before drying over $MgSO_4$ and filtered. The solvent was removed *in vacuo* and the residue was purified by silica gel chromatography [hexanes/EtOAc (9:1)].

3,4,6-Tri-*O*-acetyl-*D*-glucal (4):[15] White solid; 92% (5.00 g); mp 53-54°C; $[\alpha]_D^{20}$ -10.4 (c 1.00, MeOH); ^{1}H NMR (400 MHz, $CDCl_3$) δ 6.45 (d, *J* = 5.6 Hz, 1H), 5.32 (br s, 1H), 5.20 (dd, *J* = 5.7 Hz, 1H), 4.84–4.81 (m, 1H), 4.38 (dd, *J* = 12.0 Hz and 5.6 Hz, 1H), 4.25-4.16 (m, 2H), 2.07 (s, 3H), 2.06 (s, 3H), 2.02 (s, 3H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.2, 170.0, 169.2, 145.3, 98.6, 73.6, 67.1, 66.8, 61.0, 20.6, 20.4, 20.3.

General procedure for the synthesis of 2,3-Unsaturated *O*-Glycopyranosides (5a-f): To a 50 mL round bottomed flask containing a solution of **4** (540 mg, 2 mmol) and appropriate alcohol (2.4 mmol) in dichloromethane (10 mL) at 0^{o}C under argon was added $TeCl_4$ (11 mg, 2 mol %). The ice bath was removed and the mixture was stirred for the time indicated in Table 2. Brine (5 mL) was then added and the mixture was extracted with dichloromethane (2 x 10 mL). The combined organic phases were washed with brine (2 x 20 mL) and dried over $MgSO_4$. The solvents were removed *in vacuo* followed by purification by a flash column chromatography [hexanes/EtOAc (95:5)] to yield the corresponding 2,3-unsaturated *O*-glycopyranosides **5a**-**e**.

Prop-2-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2- enopyranoside (5a):[27] White solid; 92% (0.49 g); mp 58-59^{o}C; $[\alpha]_D^{20}$ +138.6 (c 1.00; MeOH); ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.90 (d, *J* = 10.2 Hz, 1H), 5.82 (dt, *J* = 10.2 and 1.5 Hz, 1H), 5.33 (ddd, *J* = 9.6, 3.0 and 1,5 Hz, 1H), 5.22 (br s, 1H), 4.29 (d, *J* = 2.4 Hz, 2H), 4.25 (dd, *J* = 12.4 and 5.4 Hz, 1H), 4.16 (dd, *J* = 12.4 and 2.4 Hz, 1H), 4.07 (ddd, *J* = 9.6; 5.4 and 2.4 Hz, 1H), 2.07 (t, *J* = 2.4 Hz, 1H), 2.09 (s, 3H), 2.07 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.3, 169.8, 129.3, 126.7, 92.3, 78.6, 74.4, 66.7, 64.6, 62.3, 54.6, 20.5, 20.3.

But-3-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (5b):[27] Colorless oil; 91% (0.51 g); $[\alpha]_D^{20}$ +91,3 (c 1,00; MeOH); ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.87 (d, *J* = 10.2 Hz, 1H), 5.81 (ddd, *J* = 9.6, 2.7 and 1.5 Hz, 1H), 5.28 (ddd, *J* = 9.6, 2.7 and 1.8 Hz, 1H), 5.05 (br s, 1H), 4.25-4.17 (m, 2H), 4.11 (ddd, *J* = 9.6, 5.4 and 3.0, 1H), 3.83 (dt, *J* = 16.5 and 6.6 Hz, 1H), 3.66 (dt, *J* = 16.5 and 6.6, 1H), 2.50 (td, *J* = 6.6 and 2.7 Hz, 2H), 2.08 (s, 3H), 2.06 (s, 3H), 1.97 (t, *J* = 2.7 Hz, 1H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.3, 169.8, 128.9, 126.9, 94.1, 80.5, 69.0, 66.5, 66.3, 64.7, 62.4, 20.5, 20.1, 19.6.

Pent-4-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (5c):[27] Colorless oil; 86% (0.51 g); $[\alpha]_D^{20}$ +116,7 (c 1,00; CH_2Cl_2); ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.80-5.71 (m, 2H), 5.21 (ddd, *J* = 9.6, 1.8 and 1.2 Hz, 1H), 4.94 (br s, 1H), 4.14-4.05 (m, 2H), 4.01 (ddd, *J* = 11.7, 5.1 and 2.4 Hz, 1H), 3.83-3.68 (m, 1H), 3.55-3.45 (m, 1H), 2.20 (td, *J* = 7.2 and 2.7 Hz, 1H), 2.00 (s, 3H), 1.97 (s, 3H), 1.89 (t, *J* = 2.7 Hz, 1H), 1.73 (qui, *J* = 7.2 Hz, 2H), 1.15 (t, *J* = 7.2 Hz, 1H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.0, 169.5, 128.3, 127.1, 94.0, 93.5, 82.8, 68.4, 66.3, 64.3, 62.3, 27.8, 20.4, 20.1, 14.6.

Hex-5-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (5d):[27] Colorless oil; 85% (0.53 g); $[\alpha]_D^{20}$ +95,1 (c 0.90; CH_2Cl_2); ^{1}H NMR (300 MHz, $CDCl_3$) δ 5.84 (d, *J* = 10.5 Hz, 1H), 5.78 (ddd, *J* = 10.5, 2.1 and 1.8 Hz, 1H), 5.26 (dd, *J* = 6.9 and 1.8 Hz, 1H), 4.99 (br s, 1H), 4.20 (dd, *J* = 12.0 and 5.7 Hz, 1H), 4.16 (dd, *J* = 12.0 and 2.4 Hz, 1H), 4.05 (ddd, *J* = 9.6, 5.4 and 2.7 Hz, 1H), 3.75 (dt, *J* = 9.6 and 6.6 Hz, 1H), 3.49 (dt, *J* = 9.6 and 5.7 Hz, 1H), 2.18 (td, *J* = 6.9 and 2.7 Hz, 2H), 2.06 (s, 3H), 2.04 (s, 3H), 1.93 (t, *J* = 2.7 Hz, 1H), 1.72-1.55 (m, 4H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 171.0, 170.0, 128.9, 127.7, 94.2, 83.9, 68.5, 68.0, 66.7, 65.1, 62.9, 28.5, 25.1, 20.8, 20.7, 17.9.

Dec-1-yn-3-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (5f): Colorless oil; 89% (0.65 g); $[\alpha]_D^{20}$ +149.1 (c 1.00; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3277, 2927, 2859, 1742, 1455, 1373, 1236, 1102, 1028, 735, 658; ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.92 (d, *J* = 10.0 Hz, 1H), 5.82 (ddd, *J* = 10.0, 2.4 and 2.0 Hz, 1H), 5.41 (d, *J* = 10.0 Hz, 1H), 5.18 (br s, 1H), 4.36-4.27 (m, 2H), 4.25-4.20 (m, 2H), 2.47-2.43 (m, 1H), 2.10 (s, 3H), 2.08 (s, 3H), 1.77-1.39 (m, 3H), 1.48-1.39 (m, 3H), 1.30 (br s, 6H),

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0.89 (t, *J* = 7.2 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.6, 169.9, 129.2, 127.1, 94.2, 91.3, 72.6, 68.4, 66.7, 64.7, 62.1, 35.6, 31.4, 28.8, 24.8, 24.6, 22.3, 20.6, 20.5, 13.7. HRMS (ESI, MeOH:H_2O) calcd for $C_{20}H_{30}O_6Na$ $[M + Na]^+$ 389.1940, found 389.1956.

General Procedure for the coupling reaction of Z-vinyl tellurides and α-pseudoglycosides: To a 25 mL flask under argon containing a solution of the appropriate vinyl teluride **2a**-**d** (0.5 mmol) in freshly distilled MeOH (5 mL) was added $PdCl_2$ (35 mg; 20 mol%) and CuI (40 mg; 20 mol%). The mixture was stirred at room temperature for 15 minutes and then cooled to 20°C before the addition of the appropriate α-*O*-glycoside **5a**-**d** (0.5 mmol) and Et_3N (0.3 mL; 2.2 mmol). The reaction was stirred at room temperature for the time indicated on Table 4 and the filtered through a pad of silica/celite. The mixture was extracted with EtOAc (3 x 20 mL) and the combined organic phases washed with brine (2 x 20 mL), dried over $MgSO_4$ and filtered. The solvents were removed *in vacuo* followed by purification by a flash column chromatography [hexanes/EtOAc (7:3)] to yield the corresponding enynes **6a**-**g**.

(*Z*)-5-phenylpent-4-en-2-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2- enopyranoside (6a): Yellow oil; 89% (0.16 g); $[\alpha]_D^{20}$ +122,1 (c 1.00; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3056, 2917, 2852, 1740, 1441, 1371, 1234, 1101, 1032, 736; ^{1}H NMR (300 MHz, $CDCl_3$) δ 7.73 (d, *J* = 8.1 and 1.2 Hz, 2H), 7.30-7.18 (m, 3H), 6.58 (d, *J* = 12.1 Hz, 1H), 5.85 (d, *J* = 10.2 Hz, 1H), 5.77 (ddd, *J* = 10.2, 2.4 and 1.8 Hz, 1H), 5.63 (dt, *J* = 12.1 and 2.1 Hz, 1H), 5.23-5.21 (m, 2H), 4.46 (d, *J* = 2.1 Hz, 2H), 4.16 (dd, *J* = 12.3 and 5.1 Hz, 1H), 4.08 (dd, *J* = 12.3 and 2.7 Hz, 1H), 4.03 (ddd, *J* = 9.6, 5.1 and 2.7 Hz, 1H), 2.01 (s, 3H), 2.00 (s, 3H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.3, 169.8, 139.0, 135.7, 129.3, 128.2, 128.1, 127.8, 126.9, 106.1, 92.2, 90.5, 84.5, 66.7, 64.7, 62.3, 55.5, 20.5, 20.3; HRMS (ESI, MeOH:H_2O) calcd for $C_{21}H_{22}O_6Na$ $[M + Na]^+$ 393.1314, found 393.1308.

(*Z*)-5-phenylhex-5-en-3-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2- enopyranoside (6b): Yellow oil; 86% (0.16 g); $[\alpha]_D^{20}$ +58,0 (c 0,90; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3056, 2918, 1742, 1442, 1372, 1232, 1042, 976, 735, 696; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.77 (d, *J* = 8.0 Hz, 2H), 7.28-7.25 (m, 2H), 7.21-7.18 (m, 1H), 6.51 (d, *J* = 12.0 Hz, 1H), 5.82 (d, *J* = 10.4 Hz, 1H), 5.77 (d, *J* = 10.4 Hz, 1H), 5.60 (d, *J* = 12.0 Hz, 1H), 5.23 (d, *J* = 9.6 Hz, 1H), 5.03 (br s, 1H), 4.18-4.05 (m, 3H), 3.89-3.83 (m, 1H), 3.72-3.67 (m, 1H), 2.71 (t, *J* = 6.8 Hz, 2H), 2.01 (s, 3H), 1.99 (s, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 171.0, 170.0, 138.3, 136.8, 129.6, 128.8, 128.5, 128,4, 127.9, 107.9, 94.9, 94.0, 80.5, 67.4, 67.3, 65.5, 63.2, 21.8, 21.6, 21.2; HRMS (ESI, MeOH:H_2O) calcd for $C_{22}H_{24}O_6Na$ $[M + Na]^+$ 407.1471, found 407.1463.

(*Z*)-5-phenylhept-6-en-4-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2- enopyranoside (6c): Yellow oil; 87% (0.17 g); $[\alpha]_D^{20}$ +84,2 (c 0,90; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3095, 2923, 1741, 1440, 1372, 1235, 1103, 1042, 975, 785, 736, 697; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.84 (d, *J* = 8.5 Hz, 2H), 7.34-7.26 (m, 3H), 6.57 (d, *J* = 11.6 Hz, 1H), 5.89 (d, *J* = 10.4 Hz, 1H), 5.84 (br d, *J* = 10.4 Hz, 1H), 5.68 (d, *J* = 11.6 Hz, 1H), 5.31 (d, *J* = 9.6 Hz, 1H), 5.04 (br s, 1H), 4.21 (dd, *J* = 12.8 and 5.6 Hz, 1H), 4.15 (dd, *J* = 12.8 and 2.0 Hz, 1H), 4.12-4.07 (m, 1H), 3.94-3.88 (m, 1H), 3.68-3.62 (m, 1H), 2.56 (t, *J* = 6.8 Hz, 2H), 2.08 (s, 3H), 2.07 (s, 3H), 1.95-1.87 (m, 2H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.4, 169.1, 137.3, 136.3, 128.8, 128.1, 127.8, 127.6, 127.4, 107.6, 95.0, 94.2, 79.0, 67.1, 66.6, 64.9, 28.4, 20.6, 20.4, 16.4; HRMS (ESI, MeOH:H_2O) calcd for $C_{23}H_{26}O_6Na$ $[M + Na]^+$ 421.1627, found 421.1621.

(Z)-5-phenyloct-7-en-5-yn-1-yl 4,6-Di-O-acetyl-2,3-dideoxy-α-D-erythro-hex-2-enopyranoside (6d): Yellow oil; 85% (0.17 g); $[\alpha]_D^{20}$ +75,7 (c 0,90; CH_2Cl_2); IR (KBr pellet cm-1) 3049, 2941, 1742, 1442, 1371, 1234, 1040, 907, 785, 735, 693; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.85 (d, *J* = 7.6 Hz, 2H), 7.36-7.26 (m, 3H), 6.57 (d, *J* = 12.0 Hz, 1H), 5.88 (d, *J* = 10.4 Hz, 1H), 5.84 (ddd, *J* = 10.4, 2.0 and 1.6 Hz, 1H), 5.69 (dt, *J* = 12.0 and 2.4 Hz, 1H), 5.32 (dd, *J* = 9.6 and 1.2 Hz, 1H), 5.04 (br s, 1H), 4.25 (dd, *J* = 12.0 and 5.6 Hz, 1H), 4.18 (dd, *J* = 12.0 and 2.0 Hz, 1H), 4.11 (ddd, *J* = 9.6, 5.6 and 2.0 Hz, 1H), 3.86-3.81 (m, 1H), 3.59-3.54 (m, 1H), 2.50 (td, *J* = 7.2 and 2.4 Hz, 2H), 2.10 (s, 3H), 2.09 (s, 3H), 1.84-1.67 (m, 4H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.4, 169.9, 137.1, 136.3, 128.7, 128.1, 127.8, 127.5, 107.6, 96.7, 94.1, 79.2, 67.9, 66.7, 64.9, 62.7, 28.6, 25.0, 20.6, 20.4, 19.2; HRMS (ESI, MeOH:H_2O) calcd for $C_{24}H_{28}O_6Na$ $[M + Na]^+$ 435.1784, found 435.1778.

(*Z*)-5-(3,5-dimethoxyphenyl)pent-4-en-2-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (6e): Yellow oil; 85% (0.18 g); $[\alpha]_D^{20}$ +84,4 (c 0,80; CH_2Cl_2); IR (KBr pellet cm^{-1}) 2948, 2842, 1742, 1593, 1458, 1370, 1305, 1236, 1154, 1034, 848; ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.02 (d, *J* = 2.0 Hz, 2H), 6.59 (d, *J* = 12.0 Hz, 1H), 6.43 (t, *J* = 2.0 Hz, 1H), 5.92 (d, *J* = 10.4 Hz, 1H), 5.86 (dt, *J* = 10.4 and 2.0 Hz, 1H), 5.72 (d, *J* = 12.0 Hz, 1H), 5.33 (d, *J* = 11.2 Hz, 1H), 5.28 (br s, 1H), 4.54 (br s, 2H), 4.23 (dd, *J* = 12.0 and 4.8 Hz, 1H), 4.17 (dd, *J* = 12.0 and 2.0 Hz, 1H), 4.11-4.06 (m, 1H), 3.79 (s, 6H), 2.09 (s, 3H), 2.08 (s, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.4, 169.9, 160.2, 139.1, 137.5, 129.3, 127.0, 106.3, 106.1, 101.0, 92.5, 91.3, 84.7, 66.8, 64.8, 62.4, 55.6, 54.9, 20.6, 20.4; HRMS (ESI, MeOH:H_2O) calcd for $C_{23}H_{26}O_8Na$ $[M + Na]^+$ 453.1525, found 453.1519.

(*Z*)-6-((triisopropylsilyl)oxy)tridec-4-en-2-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (6f): Yellow oil; 85% (0.25 g); $[\alpha]_D^{20}$ +110,2 (c 1,00; CH_2Cl_2); IR (KBr pellet cm^{-1}) 2931, 2862, 1746, 1458, 1372, 1233, 1037, 884, 740, 678; ^{1}H NMR (400 MHz, $CDCl_3$) δ 5.92 (d, *J* = 11.2 Hz, 1H), 5.88 (d, *J* = 10.4 Hz, 1H), 5.82 (dt, *J* = 10.4 and 2.0 Hz, 1H), 5.47 (d, *J* = 11.2 Hz, 1H), 5.33 (dd, *J* = 10.0 and 1.2 Hz, 1H), 5.25 (br s, 1H), 4.74-4.68 (m, 1H), 4.44 (d, *J* = 2.0 Hz, 2H), 4.25 (dd, *J* = 12.4 and 5.6 Hz, 1H), 4.18 (dd, *J* = 12.4 and 2.4 Hz, 1H), 4.07 (ddd, *J* = 10.0, 5.6 and 2.4 Hz, 1H), 2.10 (s, 3H), 2.08 (s, 3H), 1.66-1.58 (m, 2H), 1.53-1.44 (m, 2H), 1.31-1.21 (m, 8H), 1.05 (br s, 18H), 1.02 (br s, 3H), 0.87 (t, *J* = 6.4 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 170.3, 169.9, 147.7, 129.3, 127.0, 106.8, 92.0, 91.9, 82.7, 70.6, 66.8, 64.8, 62.5, 55.2, 37.8, 31.5, 29.4,

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28.9, 24.3, 22.3, 20.6, 20.5, 17.7, 13.7, 11.9; HRMS (ESI, MeOH:H_2O) calcd for $C_{32}H_{54}O_7Na$ $[M + Na]^+$ 601.3537, found 601.3531.

(*Z*)-6-((triisopropylsilyl)oxy)hex-4-en-2-yn-1-yl 4,6-Di-*O*-acetyl-2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (6g): Yellow oil; 84% (0.20 g); $[\alpha]_D^{20}$ +70,5 (c 0,85; CH_2Cl_2); IR (KBr pellet cm^{-1}) 2948, 2866, 1743, 1459, 1373, 1234, 1037, 683; ^1H NMR (300 MHz, CDCl3) δ 6.08 (dt, *J* = 11.1 and 6.0 Hz, 1H), 5.92 (d, *J* = 10.2 Hz, 1H), 5.82 (ddd, *J* = 10.2, 2.7 and 2.1 Hz, 1H), 5.43 (dt, *J* = 11.1 and 1.8 Hz, 1H), 5.33 (dd, *J* = 9.6 and 1.5 Hz, 1H), 5.26 (br s, 1H), 4.48 (d, *J* = 1.8 Hz, 2H), 4.45 (d, *J* = 2.1 Hz, 2H), 4.25 (dd, *J* = 12.3 and 5.1 Hz, 1H), 4.17 (dd, *J* = 12.3 and 2.7 Hz, 1H), 4.08 (ddd, *J* = 9.6, 5.1 and 2.7 Hz, 1H), 2.10 (s, 3H), 2.08 (s, 3H), 1.07-1.05 (m, 21H); ^{13}C NMR (75 MHz, $CDCl_3$) δ 170.3, 169.8, 143.6, 129.2, 126.8, 107.4, 89.3, 82.1, 66.7, 64.7, 62.3, 61.4, 55.2, 41.2, 20.5, 20.3, 17.5, 11.5; HRMS (ESI, MeOH:H_2O) calcd for $C_{25}H_{40}O_7Na$ $[M + Na]^+$ 503.2441, found 503.2435.

General Procedure for Hydrolysis of Acetyl Groups from Z-enyne pseudoglycosides: To a 50 mL flask containing the appropriate *Z*-enyne pseudoglycosides **5a**, **6a**, **6c** or **6d** (0.25 mmol) in MeOH (1 mL) was added distilled water (0.3 mL) and K_2CO_3 (69 mg; 0.5 mmol). The mixture was stirred at room temperature for 5 minutes and the solvent was removed *in vacuo*. The residue was then purified by a flash column chromatography [hexanes/EtOAc (1:1)] to yield the desired compounds **7**-**10**.

Prop-2-yn-1-yl 2,3-dideoxy-α-D-erythro-hex-2-enopyranoside (7):[28] Colorless oil; 96% (0.18 g); $[\alpha]_D^{20}$ +144,2 (c 1,00; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3370, 2948, 2875, 1490, 1263, 1028, 759; ^1H NMR (400 MHz, $CDCl_3$) δ 6.01 (d, *J* = 10.0 Hz, 1H), 5.75 (d, *J* = 10.0 Hz, 1H), 5.19 (br s, 1H), 4.30 (d, *J* = 2.0 Hz, 2H), 4.23 (d, *J* = 9.2 Hz, 1H), 3.91-3.82 (m, 2H), 3.71 (ddd, *J* = 9.2, 4.0 and 3.6 Hz, 1H), 2.70 (br s, 2H), 2.47 (t, *J* = 2.0 Hz, 1H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 133.7, 125.3, 92.5, 79.0, 74.4, 71.4, 63.6, 62.1, 54.8.

(*Z*)-5-phenylpent-4-en-2-yn-1-yl 2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (8): Colorless oil; 94% (0.27 g); $[\alpha]_D^{20}$ +154,2 (c 1.15; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3382, 2925, 2868, 1465, 1265, 1024, 886, 734, 688; ^1H NMR (400 MHz, CDCl3) δ 7.81 (d, *J* = 7.6 Hz, 2H), 7.36 (dd, *J* = 7.6 and 7.2 Hz, 2H), 7.31 (d, *J* = 7.2 Hz, 1H), 6.66 (d, *J* = 12.0 Hz, 1H), 6.01 (d, *J* = 10.0 Hz, 1H), 5.77 (d, *J* = 10.0 Hz, 1H), 5.72 (d, *J* = 12.0 Hz, 1H), 5.26 (br s, 1H), 4.53 (s, 2H), 4.24 (d, *J* = 8.8 Hz, 1H), 3.89-3.80 (m, 2H), 3.74-3.70 (m, 1H), 2.89 (br s, 2H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 139.4, 135.2, 133.8, 128.6, 128.5, 128.3, 125.8, 106.7, 92.8, 91.3, 84.9, 71.7, 64.2, 62.6, 58.0; HRMS (ESI, MeOH:H_2O) calcd for $C_{17}H_{18}O_4Na$ $[M + Na]^+$ 309.1103, found 309.1097.

(*Z*)-7-phenylhept-6-en-4-yn-1-yl 2,3-dideoxy-α-*D*-erythro-hex-2-enopyranoside (9): Colorless oil; yield 92% (0.29 g); $[\alpha]_D^{20}$ +30,5 (c 1.08; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3401, 3051, 2928, 2893, 1656, 1052, 755; ^1H NMR (400 MHz, $CDCl_3$) δ 7.84 (d, *J* = 7.2 Hz, 2H), 7.34 (dd, *J* = 7.6 and 7.2 Hz, 2H), 7.27 (t, *J* = 7.6 Hz, 1H), 6.57 (d, *J* = 12.0 Hz, 1H), 5.95 (d, *J* = 10.0 Hz, 1H), 5.74 (d, *J* = 10.0 Hz, 1H), 5.69 (dt, *J* = 12.0 and 2,4 Hz, 1H), 4.98 (br s, 1H), 4.20 (d, *J* = 9.2 Hz, 1H), 3.90 (dt, *J* = 9.2 and 6.4 Hz, 1H), 3.81 (dd, *J* = 4.0 and 3.2 Hz,2H), 3.71-3.66 (m, 1H), 3.65-3.59 (m, 1H), 2.55 (td, *J* = 6.4 and 2.4 Hz, 2H), 2.30 (br s, 2H), 1.89 (qui, *J* = 6.4 Hz, 2H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 137.5, 136.5, 133.4, 128.4, 128.2, 128.1, 126.1, 107.9, 96.5, 94.3, 79.6, 71.4, 64.1, 62.6, 28.6, 16.7.

(Z)-8-phenyloct-7-en-5-yn-1-yl)oxy 2,3-dideoxy-α-D-erythro-hex-2-enopyranoside (10): Colorless oil; yield 93% (0.29 g); $[\alpha]_D^{20}$ +40.1 (c 1.08; CH_2Cl_2); IR (KBr pellet cm^{-1}) 3420, 3061, 2929, 2895, 1640, 1052, 786, 690; ^1H NMR (400 MHz, $CDCl_3$) δ 7.85 (d, *J* = 8.0 Hz, 2H), 7.34 (dd, *J* = 8.0 and 7.6 Hz, 2H), 7.29-7.26 (m, 1H), 6.55 (d, *J* = 11.6 Hz, 1H), 5.95 (d, *J* = 10.0 Hz, 1H), 5.75 (d, *J* = 10.0 Hz, 1H), 5.68 (d, *J* = 11.6 Hz, 1H), 4.97 (br s, 1H), 4.21 (d, *J* = 9.2 Hz, 1H), 3.85 (dd, *J* = 4.8 and 4.0 Hz, 2H), 3.81-3.78 (m,1H), 3.72-3.67 (m, 1H), 3.56-3.50 (m, 1H), 2.59-2.46 (m, 2H), 2.02 (br s, 2H), 1.81-1.74 (m, 2H), 1.73-1.66 (m, 2H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 137.4, 136.6, 133.2, 128.4, 128.1 (2C), 126.3, 108.0, 97.1, 94.3, 79.5, 71.4, 68.2, 64.3, 62.8, 28.9, 25.3, 19.6.

Antiproliferative Activity: The antiproliferative activities of the synthesized compounds were evaluated in the following human cancer cells lines: HL-60 (pro-myelocytic leukemia), NCI-H292 (lung carcinoma) and MCF-7 (breast carcinoma) obtained from Rio de Janeiro Cell Bank (RJ-Brazil). All cancer cells were maintained in RPMI 1640 or DMEM medium supplemented with 10% fetal bovine serum, 2mM glutamine, 100 U/mL penicillin, 100 μg/mL streptomycin at 37°C with 5% CO_2. The cytotoxicity of all compounds was tested using the 3-(4,5-dimethyl-2-thiazolyl)-2,5-diphenyl-*2H*-tetrazolium bromide (MTT) (Sigma Aldrich Co., St. Louis, MO/USA) reduction assay. For all experiments, tumor cells were plated in 96-well plates (10^5 cells/mL for adherent cells or 3×10^5 cells/mL for leukemias). Tested Compounds (0.1–25 μg/mL) dissolved in DMSO 0.1% were added to each well and incubated for 72 h. Control groups received the same amount of DMSO. After 69h of treatment 25 μL of MTT (5mg/mL) was added, three hours later, the MTT formazan product was dissolved in 100 μL of DMSO, and absorbance was measured at 595 nm in plate spectrophotometer. The IC_{50} values and their 95% confidence intervals for two different experiments were obtained by nonlinear regression using Graphpad Prism version 5.0 for Windows (GraphPad Software, San Diego, California USA).

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Notes and references

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Electronic Supplementary Information (ESI) available: [Experimental details and characterization of the obtained compounds]. See DOI: 10.1039/x0xx00000x

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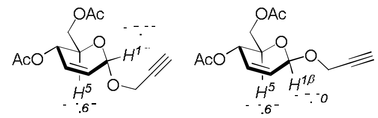
17 In compound β-**5a**, $H^{1\beta}$ and H^{5} have a *cis* relationship, whereas in α-**5a** $H^{1\alpha}$ and H^{5} have a *trans* relationship. Thus, the interaction between $H^{1\beta}$ and H^{5} would produce a positive NOE effect. This was not observed after irradiation of the signal at δ 5.22 which corresponds to the $H^{1\alpha}$ in α-**5a**.



α-**5a** β-**5a**

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Stereoselective Synthesis and Antitumoral Activity of *Z*-Enyne Pseudoglycosides

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The regio- and stereoselective synthesis of lower structural complexity analogues of enynes and enediyne natural products is described. Screening against tumor cell lineages indicated that the synthesized compounds are promising intermediates for the synthesis of an array of more potent target structures.